

04	Vertical lift force labeled <i>L</i> of the same length as <i>W</i> ✓ Horizontal drag force labeled <i>D</i> of the same length as <i>T</i> ✓ The pairs of forces are co-linear OR are in positions to suggest that there is no net moment. ✓	3	Allow one mark only for freehand sketches if the lengths and directions are approximately correct by eye
Total		3	

Question	Marking guidance	Mark	Comments
	Draws gradient as a tangent to the curve on the graph ✓ Extracts at least one piece of data from the graph that is used to		Candidates who use a small triangle can access the second and third marks only

06	Clear attempt to construct a moments equation ✓ Two correct terms seen: (2×700), (2.5×470), $(4F_s)$ ✓ 640 or 644 N ✓	3	Minimum requirement is a recognisable force $\times$ distance on either side of an equals sign Accept as correct terms 1400 and/or 1175 Expect to see $(2 \times 700) + (2.5 \times 470) = 4 F_s$
	(electron) antineutrino or $\bar{\nu}$ or $\bar{\nu}$ ✓	1	Accept misspellings when intention is clear but not

15	B
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Question	Marking guidance	Mark	Comments
03.1	Positron ✓	1	Condone spelling error if meaning is clear Allow antielectron or e^+ or β^+

Question	Marking guidance	Mark	Comments
03.2	Energy of photon needs to provide at least the <u>rest</u> masses OR at least the <u>rest</u> energy ✓ of the electron <u>and</u> positron / of (both) particles / of particle and antiparticle ✓	2	Allow particles or products MP2 is contingent on MP1 being awarded.

Total		3	
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Question	Marking guidance	Mark	Comments
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13	B	$F = T \times \sin \theta$
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14	B	larger in P1 than in P2	larger in P1 than in P2
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Question	Answers	Additional comments/Guidelines	Mark	AO
10.1	$Mga + wb/2 = Fb$ seen ✓		1	AO2

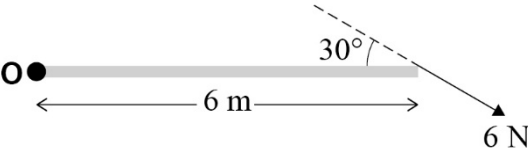
Question	Answers	Additional comments/Guidelines	Mark	AO
10.2	Good best-fit line ✓	Look for intercept of between 1.0 and 1.2 (N) The line should accurately follow the trend of the points with an even scatter of points either side of the line.	1	AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
10.3	Large triangle drawn (or implied with values taken from the line of best fit) to find the gradient ✓ Gradient in the range 8.0 to 8.3 ✓ Correct calculation leading to M in the range 0.65 to 0.68 (kg) ✓	Alternative for MP1 and MP2 using the equation for the line: MP1 choose points from large distance apart from the line of best fit. MP2 Two simultaneous equations correct	3	1 × AO2 1 × AO3 1 × AO4

Question	Answers	Additional comments/Guidelines	Mark	AO
10.4	Intercept accurately read from graph in the range of 1.0 to 1.2 (N) OR doubles intercept to find w ✓ Zero error in range -0.2 to -0.6 (N) ✓	Allow consistent of point on the line. Accept 1 dp only	2	1 × AO3 1 × AO4

Question	Answers	Additional comments/Guidelines	Mark	AO
10.5	No effect because gradient is not affected by zero error OR the same error affects every measurement of F ✓		1	AO4
Total			8	

Question	Key	Answer	AO
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13	B	 The diagram shows a horizontal beam of length 6 m pivoted at point O. A force of 6 N is applied at the other end at an angle of 30 degrees to the beam.	AO3
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Question	Answers	Additional comments/Guidelines	Mark	AO
09.1	the force required is less at A / the moment created is greater at A (for the same force) AND statement of moment = force $\times$ (perpendicular) distance to pivot ✓		1	AO1

Question	Answers	Additional comments/Guidelines	Mark	AO
09.2	they are not a couple because the forces are not equal in magnitude OR the forces are not antiparallel/opposite OR a net force is exerted ✓		1	AO1

Question	Answers	Additional comments/Guidelines	Mark	AO
09.3	(use of $T = Fd$ to give) 3.8(2) (Nm) ✓		1	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
09.4	use of Pythagoras to get magnitude = 2.7 (N) ✓ use of $\tan^{-1}(1.8/2.0)$ or vice versa ✓ to give 42° ✓	alternative for scale diagram method: a correct vector triangle of suitable size seen ✓ magnitude 2.5 to 2.9 N ✓ angle 40 to 45° ✓	3	AO3 $\times$ 2 AO2 $\times$ 1

Question	Answers	Additional comments/Guidelines	Mark	AO
09.5	arrow from central shaft going down and slightly to the right with the angle labelled ecf from 09.4 ✓	allow ecf for any direction in the bottom quadrant that matches their 09.4	1	AO3
Total			7	

Question	Answers	Additional comments/Guidelines	Mark	AO
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11	D	N s	AO1
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12	B	<table border="1"><thead><tr><th>Category</th><th>Mean Length (mm)</th><th>Lower Bound (mm)</th><th>Upper Bound (mm)</th></tr></thead><tbody><tr><td>A</td><td>13.0</td><td>12.0</td><td>14.0</td></tr><tr><td>B</td><td>12.4</td><td>10.4</td><td>14.4</td></tr><tr><td>C</td><td>12.4</td><td>11.4</td><td>13.4</td></tr><tr><td>D</td><td>12.0</td><td>11.0</td><td>13.0</td></tr></tbody></table>	Category	Mean Length (mm)	Lower Bound (mm)	Upper Bound (mm)	A	13.0	12.0	14.0	B	12.4	10.4	14.4	C	12.4	11.4	13.4	D	12.0	11.0	13.0	AO4
Category	Mean Length (mm)	Lower Bound (mm)	Upper Bound (mm)																				
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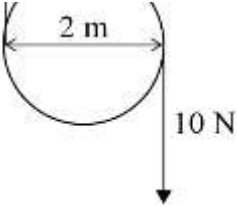
13	C	26 N	AO2
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Question	Answers	Additional comments/Guidelines	Mark	AO
06.1	There is no resultant moment (about any point) ✓ There is no resultant force ✓		2	AO1

Question	Answers	Additional comments/Guidelines	Mark	AO
06.2	Uses $W = mg$ to determine weight of skateboard (47 N) ✓ Adds to the weight of the child and halves to give 204 (N) ✓	Allow 200 (N) (2sf) but not 203 (N)	2	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
06.3	$F_A = 30$ (N) or $F_B = 330$ (N) ✓ Takes moments about P ACW moment = $330x$ ✓ CW moment = $(4.8 \times 9.81) 32.5\cos 15 + (30 \times 53\cos 15)$ ✓ Uses POM to give $x = 9.1$ (cm) ✓	Allow mp2 and mp3 to be combined Allow mp2 and mp3 in terms of F_A and/or F_B Accept answer in m if unit correct	4	AO2 $\times$ 2 AO3 $\times$ 2
Total			8	

Question	Answers	Additional comments/Guidelines	Mark	AO
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24	A		
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Total 14 marks

Question	Answers	Additional comments/Guidelines	Mark	AO
10.1	Equates the clockwise moment to the anticlockwise moment and correct rearrangement ✓	Expect to see $Mgs = mg(d-s)$	1	AO1

Question	Answers	Additional comments/Guidelines	Mark	AO
10.2	Correct plotting of values as x or + ✓	(6.71,0.40), (8.00,0.50), (9.35,0.60) Condone $\odot$ Allow $\pm$ half a grid square horizontally	1	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
10.3	Line of best fit with equal number of crosses either side of line ✓		1	AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
10.4	Correctly reads their M -intercept ✓	Expect to see 0.14–0.16 (kg)	1	AO1

Question	Answers	Additional comments/Guidelines	Mark	AO
10.5	Points taken from large gradient triangle ✓ Gradient in the range 0.079–0.083 ✓ Divides their gradient by their answer to Question 10.4 ✓	At least 5 cm horizontally expect to see d in the range 0.50 to 0.56 (m)	3	1 × AO2 1 × AO3 1 × AO4
Total			7	

Question	Answers	Additional comments/Guidelines	Mark	AO
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Question	Answers	Additional comments/Guidelines	Mark	AO
02	Idea that they are equal in magnitude and opposite in direction ✓ They are separated by a perpendicular distance and are coplanar ✓		2	AO1
Total			2	

Question	Answers	Additional comments/Guidelines	Mark	AO
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15	D	0.51 N	AO2
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20	C	12 N 16 N 20 N	AO2
		R	

Question	Answers	Additional comments/Guidelines	Mark	AO
02.1	(A quantity that has) both magnitude and direction ✓	Allow 'size' for magnitude	1	AO1

Question	Answers	Additional comments/Guidelines	Mark	AO
02.2	Correct vector quantity AND correct scalar quantity ✓	e.g. displacement and distance, velocity and speed, etc	1	AO1

Total			2	
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Question	Answers	Additional comments/Guidelines	Mark	AO
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Question	Answers	Additional comments/Guidelines	Mark	AO
07.1	Calculation of initial velocity of target and arrow from conservation of energy ✓ Calculation of initial momentum of target and arrow OR Momentum after = momentum before ✓ $u = 23.4 \text{ (m s}^{-1}\text{)} \checkmark$	Expect to see $u = \sqrt{2gh} = 1.7155\dots$ $2.22\dots \text{ N s}$ $2.22 = 0.095 \times u$ (or correct momentum equation) 23.3 if 9.8 used for g Do not allow use of suvat in MP1 but allow ecf in MP2 and MP3 if used	3	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
07.2	Height is greater, with any three from: ✓✓✓ - change in momentum / impulse of arrow will be greater (therefore) change in momentum / impulse of target will be greater to conserve momentum - kinetic energy of target after impact will be greater OR initial velocity / speed of the target is greater - gravitational potential energy of target will be greater	Alternative for mp1 and mp2: - arrow will have negative momentum - target must have more positive momentum to conserve momentum.	3	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
07.3	Use of $E_k = \frac{1}{2}mv^2$ (91.96 J) ✓_{1a} Equates their kinetic energy with work done (Fs) ✓_{2a} Use of $v^2 = u^2 + 2as$ to determine acceleration (80 667 m s⁻²) ✓_{1b} Use of $F = ma$ with their a ✓_{2b} Use of $s = (u + v) \frac{t}{2}$ to determine Δt (5.45×10^{-4} s) ✓_{1c} Use of $F = \frac{\Delta mv}{\Delta t}$ with their Δt ✓_{2c} (-)7700 (N) ✓₃	(7663 N)	3	AO2
Total			9	

Question	Answers	Additional comments/Guidelines	Mark	AO
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21	C	$mg \sin \theta$ $+y$	AU2
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23	B	0.32 m	AO2
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